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MZ 505
Inventory Systems

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Session 06:
The Newsvendor Model

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Overview

- Demand uncertainty is one of the key challenges in operations and supply chain management.
- It creates numerous problems, e.g. how much
  - Capacity should we maintain
  - Of different products should we produce
  - Inventory should we hold
- The well-known “newsvendor” model is fundamental in addressing these issues.
  - We will focus on its application in inventory/production decisions.

Acknowledgement is hereby given to Prof. Richard Pibernik for providing his advice and some material for this slideshow.

The newsvendor problem



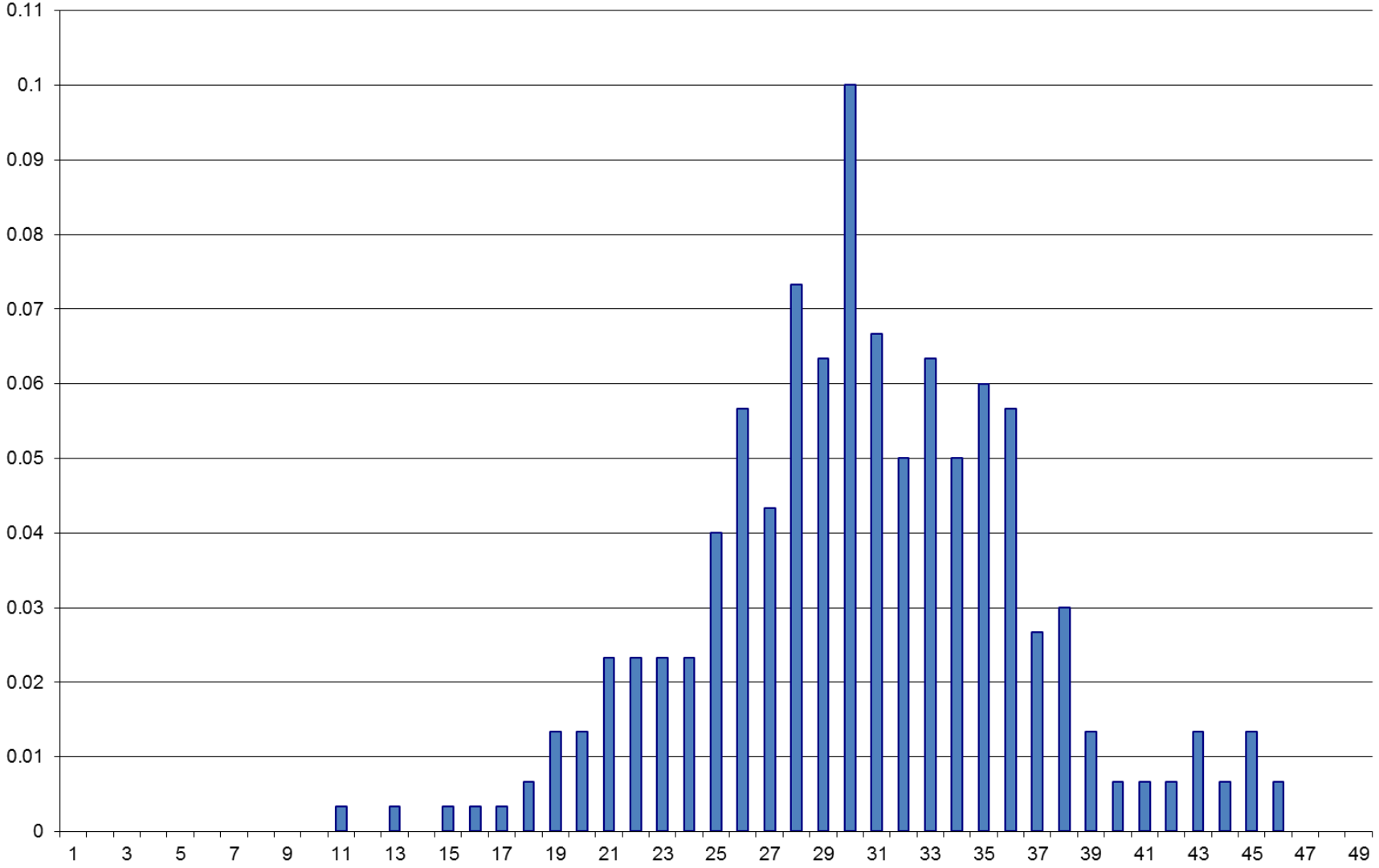
Demand uncertainty

- Demand “certainty”: We know exactly how much the customers will want to buy in a certain time period (e.g. day, month, year, etc.).
- Demand uncertainty means that customer demand in a certain time period is a **random variable** (let’s call it D).
- Beforehand we don’t exactly know the **realization** (call it d) of this random variable.
- It is usually not the case that we don’t know anything about the demand; we usually have some information, e.g. from
  - Sales history
  - Intuition, experience, estimates, etc.
  - Other sources (e.g. how much are others selling)

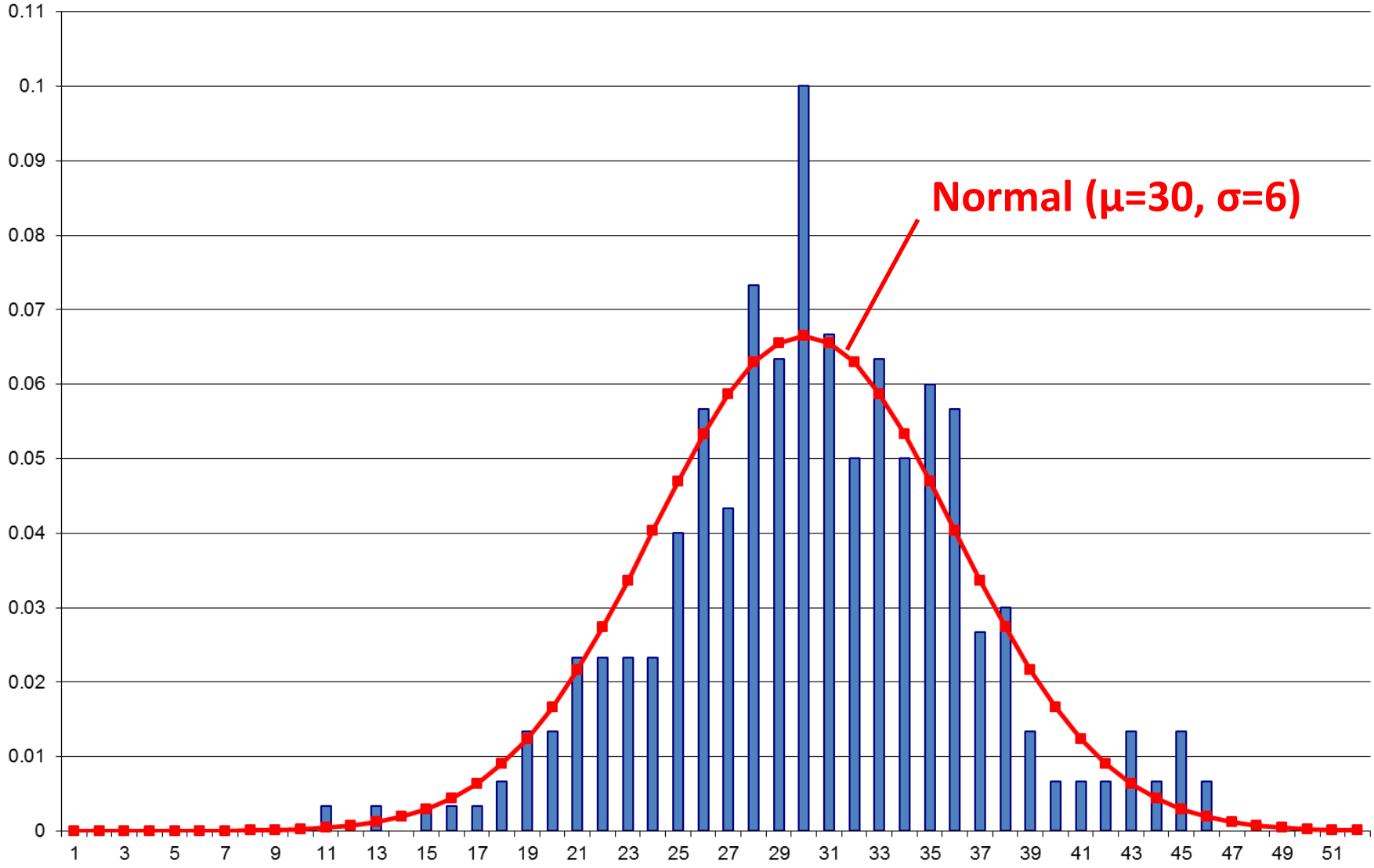
Demand uncertainty: A simple intuitive example

- Consider, for example, a bakery that produces bread in the morning and sells it during the day.
  - Before the day starts, the bakery knows there will be an uncertain demand D during the day.
  - However, only at the end of the day it knows the true demand, d (we say “realization”).
- Assume that the bakery knows the demand over the last 300 days.
  - This demand includes the number of breads sold and the ones not sold.
  - Demand is independent of the day, month, etc. and there is no trend.
- How can we “**characterize**” D? Let’s take a look at the data in Excel!

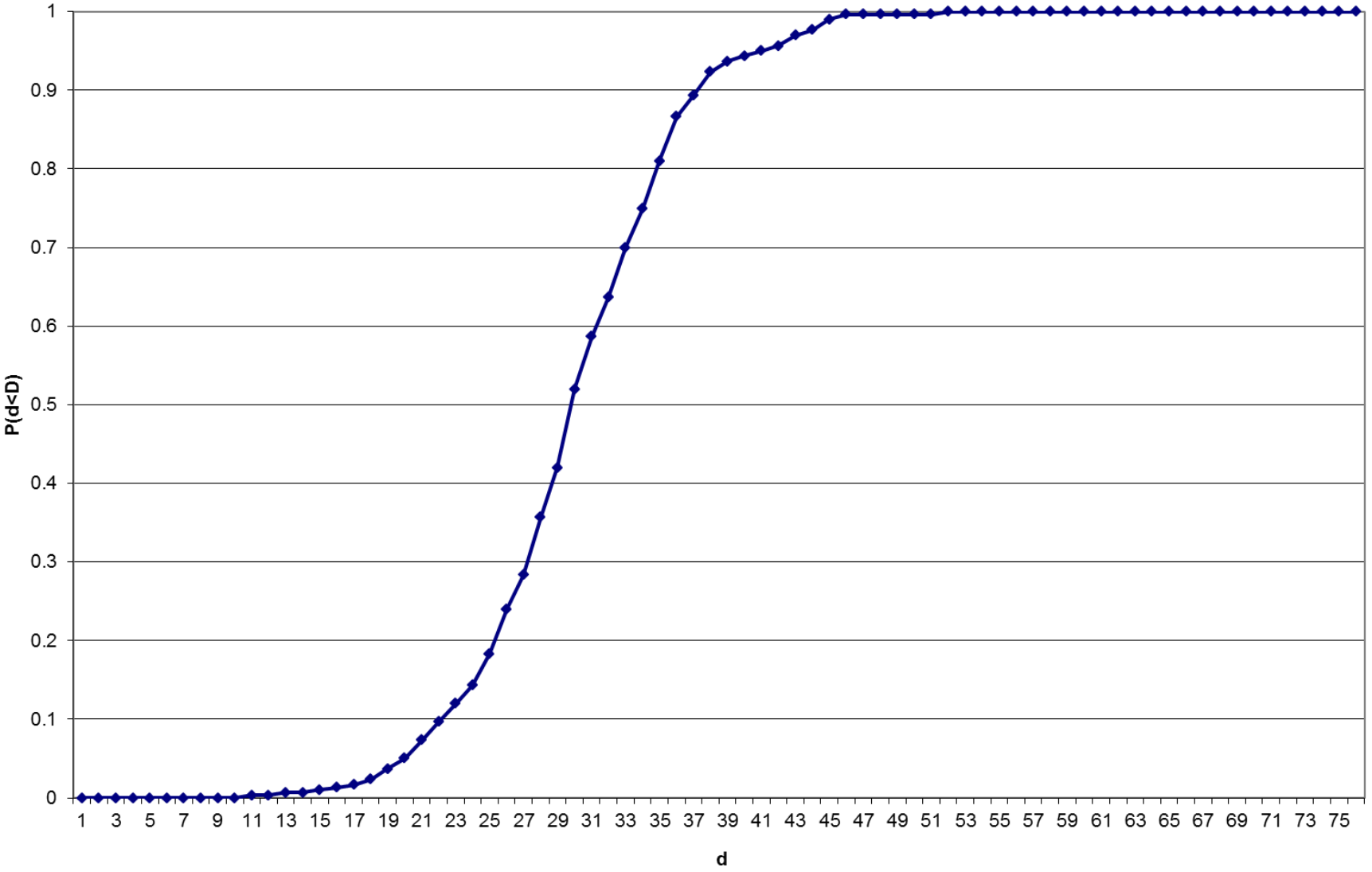
Characterizing demand: Histogram



Characterizing demand: PDF fitting

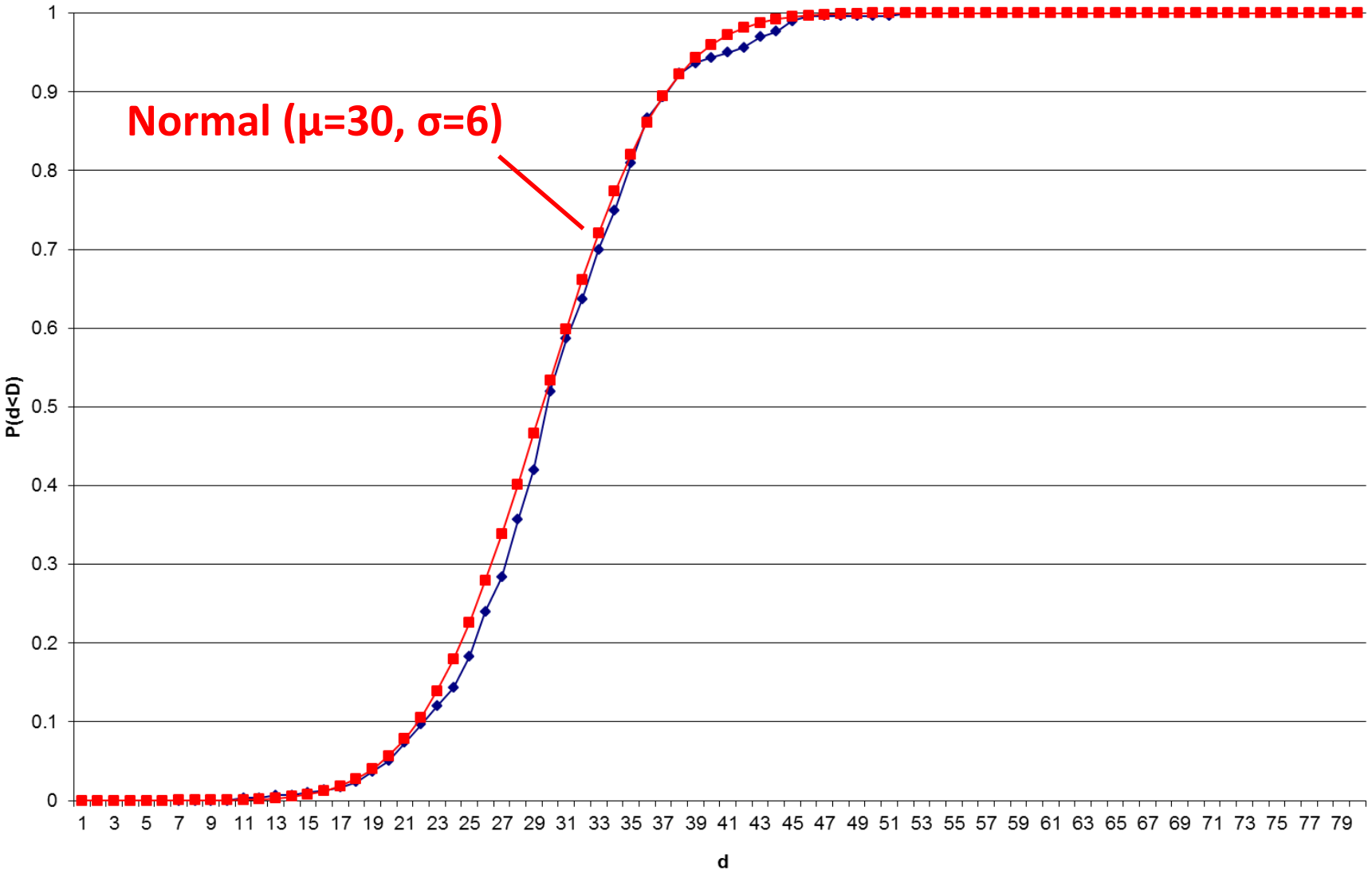


Characterizing demand: CDF



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Characterizing demand: CDF fitting



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Approximating the demand distribution

- Based on our historical data we can approximate the distribution of D (not only moments but also the shape).
- From our example you should understand:
  - How we can determine the (empirical) probability that the uncertain demand D is equal to some realization d; i.e. P(D=d).
  - How we can approximate the distribution D by means of a theoretical distribution.
    - ✓ We saw that the demand for bread can be approximated by a Normal distribution with Mean 30 and Standard Deviation of 6 – we write D ~ N(30,6).
  - The meaning of the cumulative distribution function CDF (we will need this to carry on with our problem).

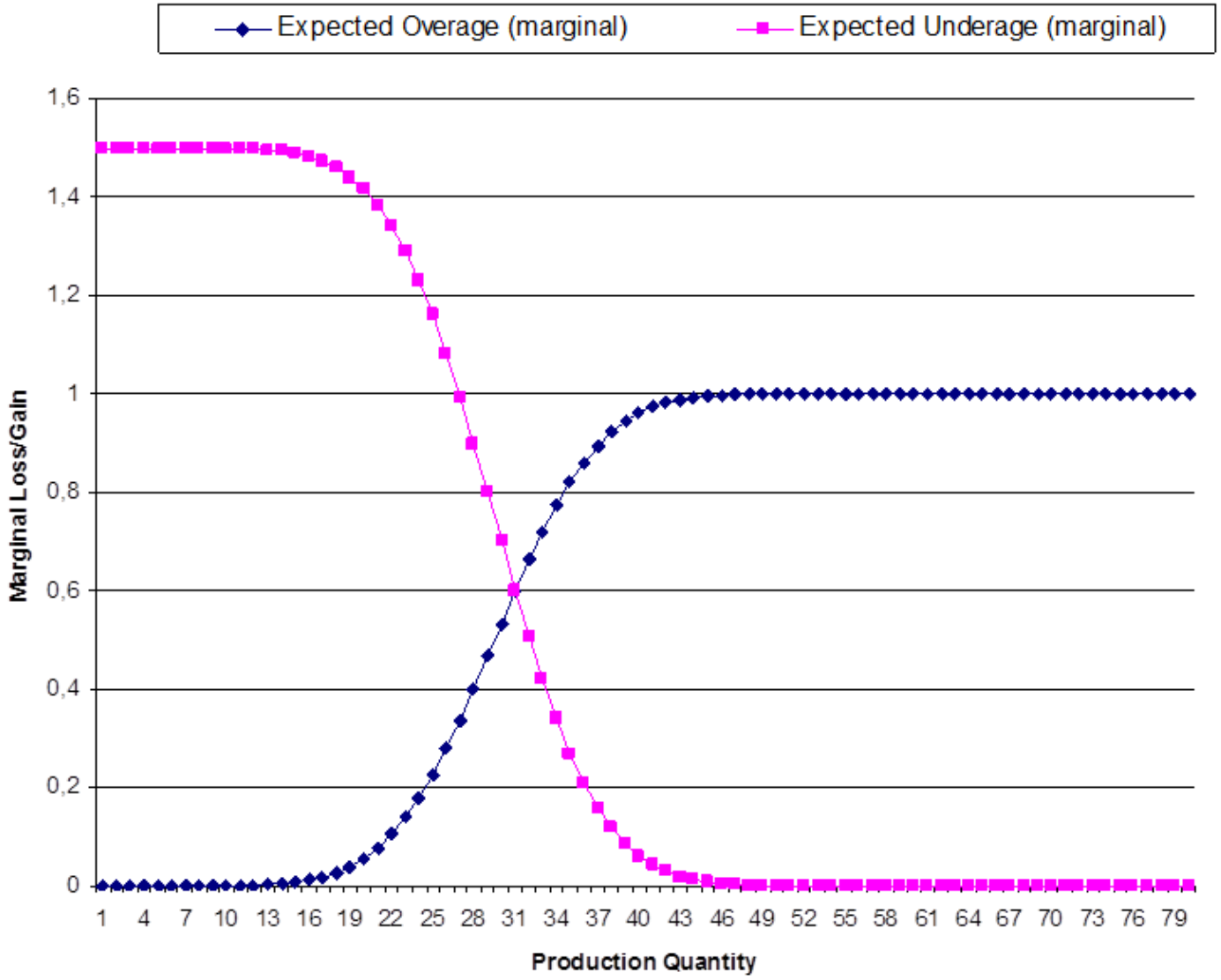
The optimal production quantity (1)

- For the bakery, the most important question is:
  How much bread should be produced every day?
- Based on our prior analysis, we assume that D ~ N(30,6).
- Let‘s keep it simple and assume the cost of producing one bread loaf is €1 and the sales price is €2.5.
- Assume that any excess bread at the end of the day is donated to some charity organization.
- How can we calculate the optimal number of bread loafs to be produced on every day?
- Let‘s do some simple computations in Excel

The optimal production quantity (2)

- From our experiment we can see that the optimal number of bread loafs depends on the production cost and the sales price.
- We can calculate the so-called **"critical ratio"** from "overage" and "underage" cost.
- **Overage cost (C_o)**: The cost of making one unit more than what you would have ordered had you known the true demand.
  - E.g., suppose you had left over breads (i.e., you over produced)
  - C_o is the increase in profit you would have made, had you produced one unit less; in our case €1
- **Underage cost (C_u)**: The cost of making one unit less than what you would have ordered had you known the true demand.
  - E.g., suppose you had lost sales (i.e., you under produced)
  - C_u is the increase in profit you would have made, had you produced one more unit; in our case 2,5-1=1,5€

The optimal production quantity (3)



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The "newsvendor" solution

- The optimal production quantity is given where expected benefit of making one more unit is equal to the expected loss from making one more unit, i.e. when

$$C_o \times F(Q) = C_u \times (1 - F(Q))$$

- Rearrange terms in the above equation → $$F(Q^*) = \frac{C_u}{C_u + C_o}$$
- This reads: The probability that the demand realization d is less than or equal to the optimal production quantity Q* must be equal to the **critical ratio** of underage and overage costs.
  - I.e., just by looking at the CDF we can determine Q*
  - We often use the term "Type I - service level" (often denoted by "*α*") if we talk about the probability of not "stocking-out" in a certain time period.

A different way to write F(Q) = α*

- Assume, as before, that demand D follows a normal distribution with mean μ and standard deviation σ.
- Remember from your statistics courses: any normally distributed random variable D can be related to the standard normal Z:

$$Z = \frac{D-\mu}{\sigma}$$

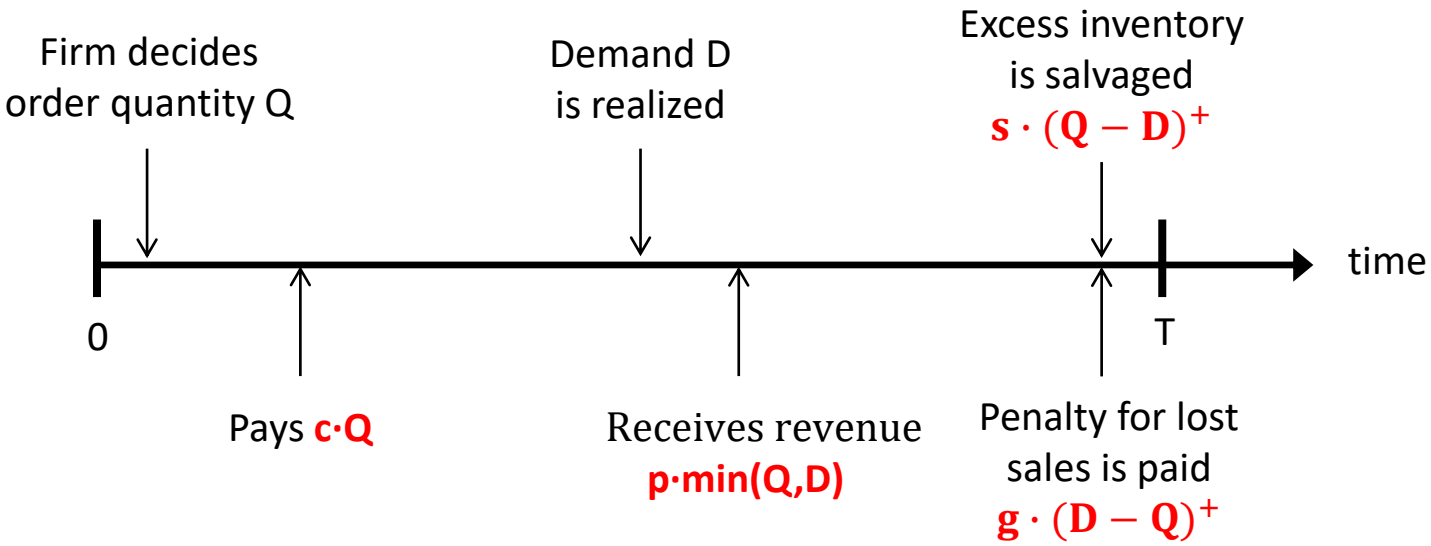
- Z is a random variable that is normally distributed with mean=0 and standard deviation=1
- Rearranging terms gives us $D = \mu + Z\sigma$
- Then, we can rewrite $F(Q^*) = a$ to $Q^* = F^{-1}(a) = \mu + z(a)\sigma$
  - Where z(*a*) is the inverse of the standard normal at *a*, $\Phi^{-1}(a)$
  - In excel the command is normsinv(*a*)

The Newsvendor Model: Formal presentation

- Newsvendor is a single period, stochastic demand model (i.e., demand D is a random variable).
  - Let denote by f(D) and F(D), respectively, the PDF and CDF of demand over the selling season.
- Sequence of events:
  - At the beginning of the period the firm decides an order quantity Q.
  - During the period, demand is realized and served to the extend possible.
  - At the end of the period, lost sales (if any) incur a penalty cost.
  - At the end of the period, excess inventory (if any) is salvaged.
- The **objective** of the firm is to **decide an order** (or production) quantity that **maximizes expected profit** over the selling horizon.

The Newsvendor Model: Basic setting

- The problem parameters are as follows:
  - D: demand
  - Q: order quantity
  - p: unit revenue
  - c: unit cost
  - s: salvage value per unit leftover
  - g: goodwill cost per unit short



$(x)^+ = \max(0, x)$

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Newsvendor model: Objective function

- Objective function: choose Q to maximize the total **expected profit**

$$\pi(Q) = E[\underbrace{p \cdot \min(Q, D)}_{\text{sales}} + \underbrace{s \cdot (Q - D)^+}_{\text{salvaged}} - \underbrace{g \cdot (D - Q)^+}_{\text{short}} - \underbrace{c \cdot Q}_{\text{purchasing}}]$$

- By using the following transformations

$$\min(Q, D) = D - (D - Q)^+$$

$$Q = D - (D - Q)^+ + (Q - D)^+$$

- We get:

$$\pi(Q) = \underbrace{(p - c)E[D]}_{\text{Profit with perfect demand information}} - E[\underbrace{(p + g - c)}_{\text{Underage cost (marginal)}} \cdot (D - Q)^+ + \underbrace{(c - s)}_{\text{Overage cost (marginal)}} \cdot (Q - D)^+]$$

Newsvendor model: Solution

- The profit maximization problem is equivalent to minimizing of:

$$C(Q) = E[C_u \cdot (D - Q)^+ + C_o \cdot (Q - D)^+]$$

where $C_u = p + g - c$ and $C_o = c - s$

- This is formally written as:

$$\min_Q C(Q) = C_u \int_Q^{\infty} (x - Q) f(x) dx + C_o \int_{-\infty}^{Q} (Q - x) f(x) dx$$

- To determine the minimum of the expected costs we differentiate the expected cost function with respect to Q:

$$\frac{dC(Q)}{dQ} = C_o F(Q) - C_u \big(1 - F(Q)\big) = 0$$

- As the cost function is convex, it follows that:

$$F(Q^*) = Pr(D \leq Q^*) = \frac{C_u}{C_u + C_o} = \frac{p - c + g}{p - s + g} = \alpha$$

Generalization

- To minimize the expected total cost of underage and overage, choose Q such that the probability that we don't have lost sales (i.e., demand is Q or lower) equals the critical ratio.
- **Realize that this rule does not hinge upon the assumption that demand is normally distributed!**
  - You can employ this decision rule for any demand distribution (that you generate from historical data).
  - You just have to generate the pdf and cdf (more on this in our demand forecasting sessions).
- Let's take a look at a final example.

Example

- Assume that the critical ratio $C_u/(C_u + C_o)=0.6$
- We have some (different) past sales data based on which we have determined the following (discrete) probabilities P(D=d) and the CDF F(D):

D	22	24	26	28	30	32	34	36
Probability	0.05	0.1	0.15	0.2	0.2	0.15	0.1	0.05
$F(D)$	0.05	0.15	0.3	0.5	0.7	0.85	0.95	1



- Given our critical ratio of 0.6, our production quantity should be 30 units.
  - If the critical ratio falls between two values in the CDF, choose the greater… this is called the **round-up rule**.
  - The same holds when a desired probability falls between two values in the z-table.

Key Takeaways

- Steps to determine the optimal inventory/production quantity:
  - Determine demand distribution (forecast, forecast error).
  - Determine optimal service level (critical ratio).
  - Combine the two in the NV formula to determine optimal quantity.
- Optimal service level is determined by how much it costs to have one unit too little (underage) relative to the cost of having one unit too much (overage):
  - What determines underage? → lost sales, penalties, loss of goodwill, etc.
  - What determines overage? → inventory holding cost, obsolescence, depreciation
- Optimal service level is independent of demand distribution, but demand distribution (i.e., its standard deviation) drives system cost (HW P4).

Readings

- Chapter 12 (sections 12.1, 12.2, and 12.4) from Cachon & Terwiesch, 3rd edition

Note: Section 12.3 will be covered in our "demand forecasting" sessions

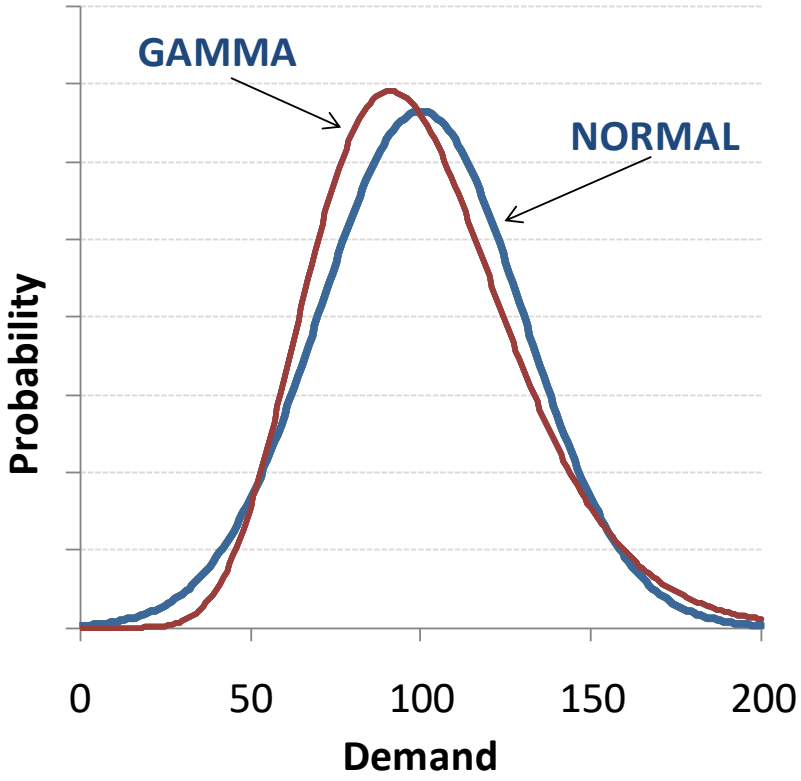
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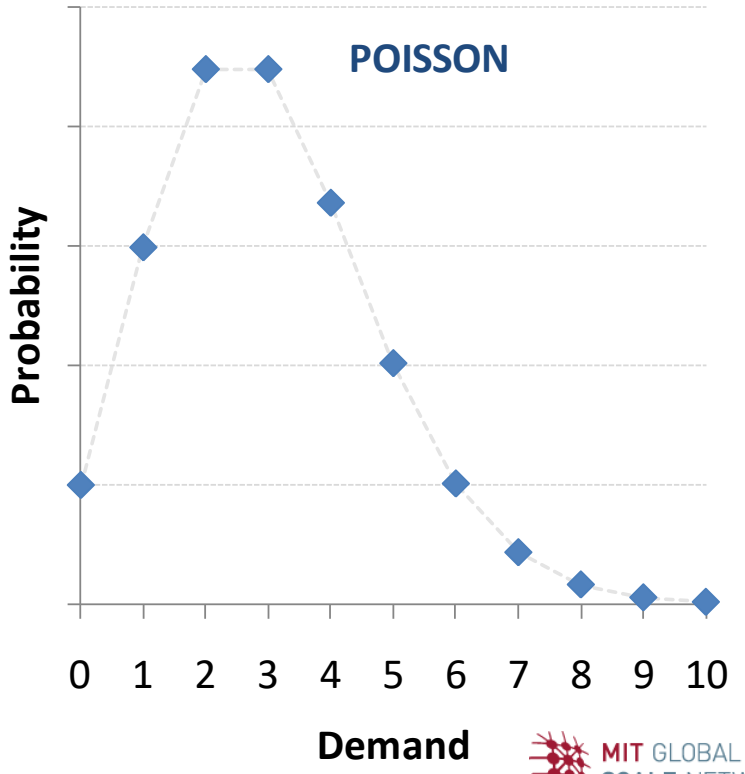
A REVIEW OF DEMAND MODELS

Adapted from Cachon/Terwisch (2013)

What is a demand model?

- A demand model specifies what demand outcomes are possible and the probability of these outcomes.
- Traditional distributions from statistics can be used as demand models:
  - e.g., the normal, gamma, Poisson distributions





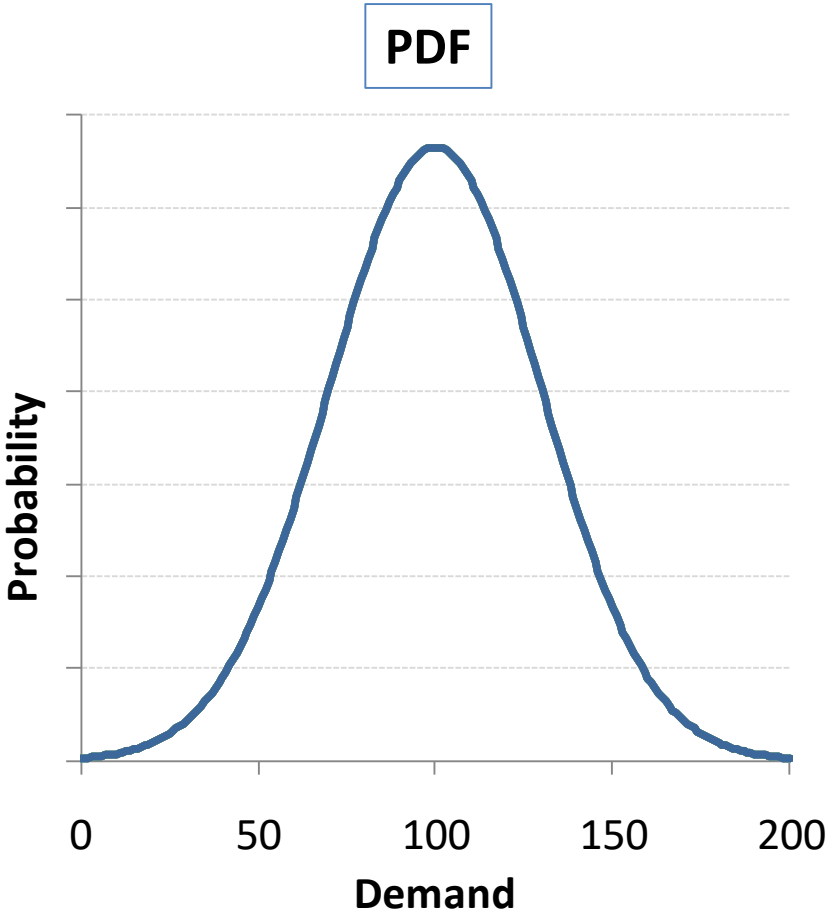
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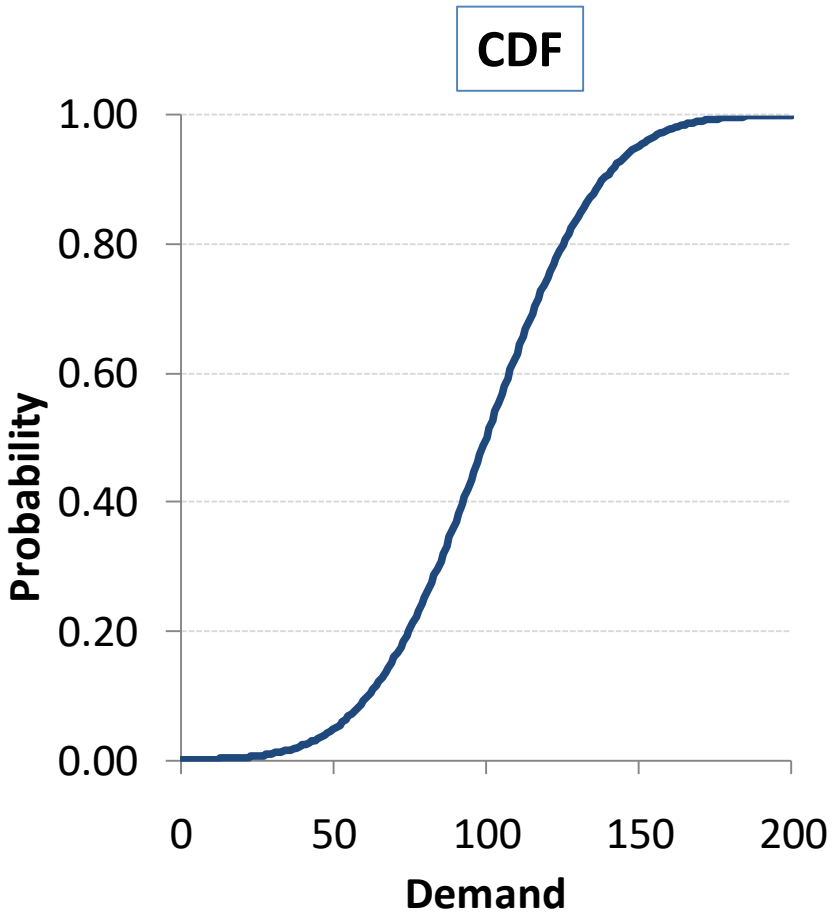
Distribution and density functions

- The **probability density function (PDF)** gives the probability of a particular outcome occurring
  - E.g. the probability that demand will be exactly 5 units
- The **cumulative distribution function (CDF)** gives the probability the outcome will be a particular value or smaller
  - E.g. the probability demand will be 5 or fewer units
- Both can be used to describe a demand model but we will more often work with distribution functions
- Density functions can take all kinds of shapes (they don't always have to look like a bell curve)
- Distribution functions always start low and increase towards 1

Distribution and density functions (cont.)

- Two ways to describe the same demand model:





Random demand and notation

D : demand during the selling season – a random variable

P(x): probability distribution function for discrete D
(gives the probability that the demand equals x)

$F(b) = P[D \leq b] = \sum_{x=0}^{b} P(x)$: CDF for discrete D

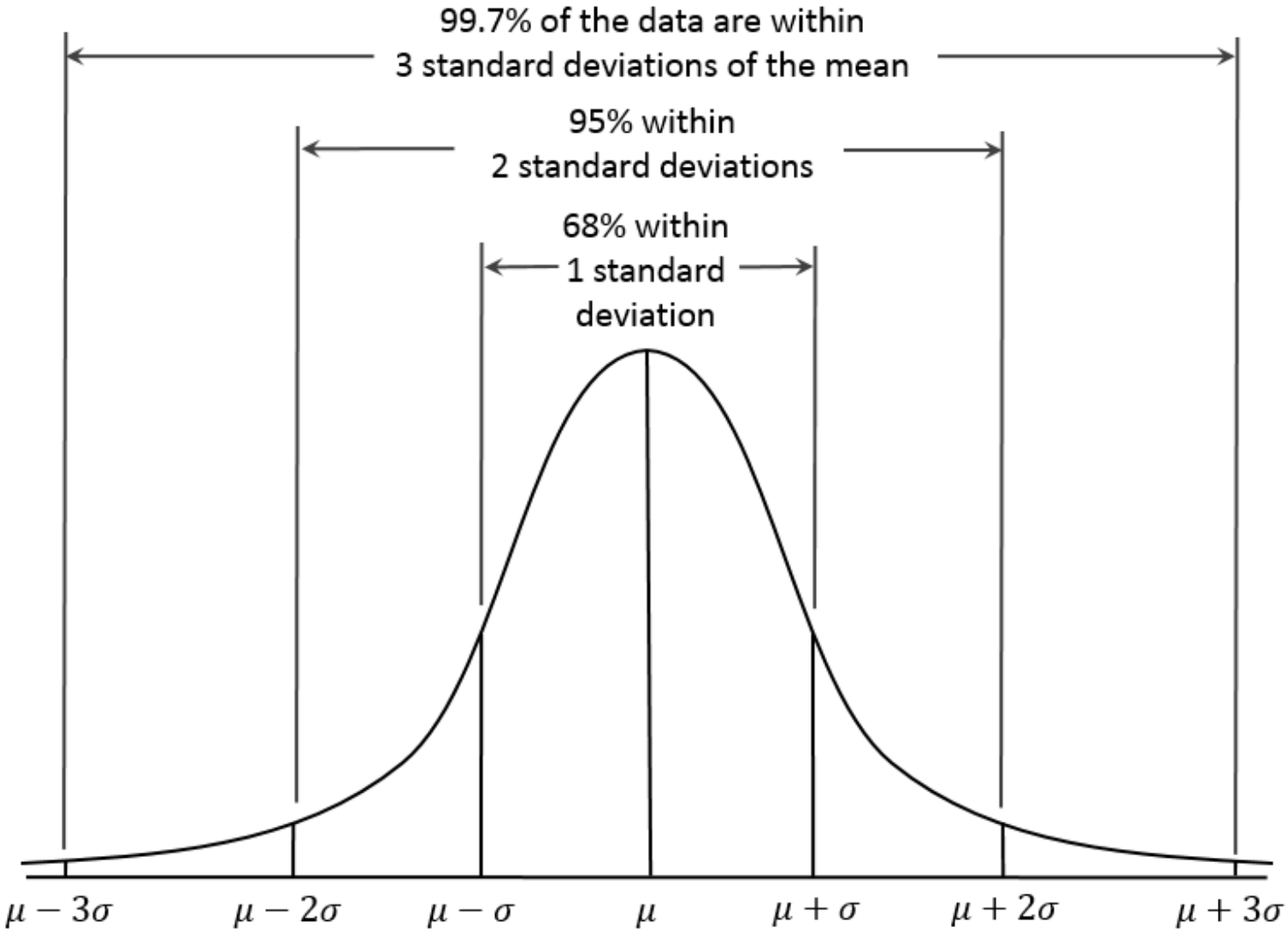
f(x) : PDF for continuous D

$F(b) = P[D \leq b] = \int_{-\infty}^{b} f(x)dx$: CDF for continuous D

$\mu = \int_{-\infty}^{\infty} xf(x)dx$: mean

$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx$: variance

Normal distribution



Normal and Standard Normal distributions

- All normal distributions are characterized by two parameters, mean (μ) and standard deviation (σ)
- All normal distributions are related to the **standard normal** that has mean μ = 0 and standard deviation σ = 1
  - We use $\varphi(x)$ and $\Phi(x)$ to denote respectively the PDF and CDF of the standard normal distribution
- For example, let "x" be the order quantity, and (μ, σ) the parameters of the normal distribution used to model demand:
  - $F(x) = P[D \leq x] = \Phi(z)$, where $z = \frac{x-\mu}{\sigma}$
- Look up z in the *Standard Normal Distribution Function Table* to get the desired probability or use Excel:
  - $\Phi(z)$ = normsdist (z)

The Standard Normal Distribution Function Table

Values of z on the outside

Standard Normal Distribution Function Table (continued), $\Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621

Probabilities in the inside

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Standard Normal Distribution Table: Example

How to start with a value of z and find the corresponding probability?

Q: What is the probability the outcome of a standard normal will be z = 0.28 or smaller?

Look for the intersection of the fourth row (with the header 0.2) and the ninth column (with the header 0.08) because 0.2+0.08 = 0.28, which is the z we are looking for.

A: The answer is 0.6103, which can also be written as 61.03%

Standard Normal Distribution Function Table (continued), $\Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
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0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224

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Standard Normal Distribution Table: Example

How to start with a probability and find the corresponding value of z?

Q: For what z is there a 70.3% chance that the outcome of a standard normal will be that z or smaller?

Find the probability inside the table (if between two values **round up**)

Add the row and column headers for that probability

A: the answer is z = 0.54

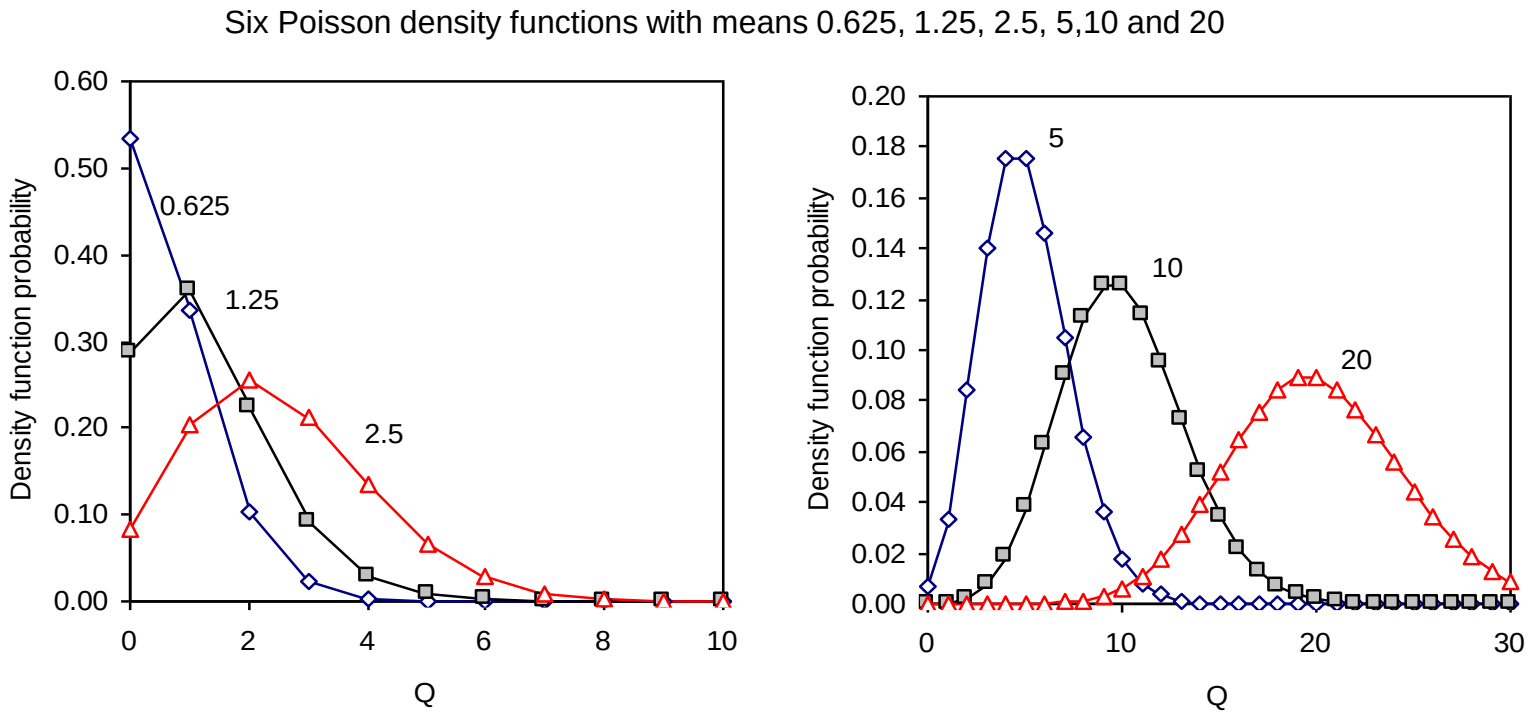
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0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224

round-up rule

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Poisson distribution



- Defined only by its mean (standard deviation = square root(mean))
- Does not always have a “bell” shape, especially for low demand
- Discrete distribution function: only non-negative integers
- Good for modeling demands with low means (e.g., less than 20)
- If the inter-arrival times of customers are exponentially distributed, then the number of customers that arrive in a given interval of time has a Poisson distribution

Poisson distribution: Example

- EcoTea plans to sell a gift basket of Tanzanian teas through its specialty stores during the Christmas season
- Basket price is $55, purchase cost is $32 and after the holiday season left over inventory will be cleared out at $20
- Estimated demand at one of its stores is Poisson with mean $\lambda = 4.5$
- One order is made for the season
- What is the optimal order quantity?
  - ✓ C_o=32-20=12; C_u=55-32=23; $C_u / (C_u + C_p) = 23 / 35 = 0.6571$
  - ✓ Use round up rule, so order 5 baskets

Poisson Distribution Function Table

		Mean	
S	...	4.25	4.5
0		0.01426	0.01111
1		0.07489	0.06110
2		0.20371	0.17358
3		0.38621	0.34230
4		0.58012	0.53210
5		0.74494	0.70293
6		0.86169	0.83105

$$\sum_{x=0}^{Q} P(x) = \sum_{x=0}^{Q} \frac{\lambda^x e^{-\lambda}}{x!}$$

In excel: poisson.dist(S; λ; true)

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